



VR basics

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1. Einleitung

2. Zielsetzung

3. Methodik

4. Ergebnisse

5. Diskussion

6. Fazit

7. Literaturverzeichnis

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9. Zusammenfassung

10. Schlusswort





Extended Reality (XR)



Extended Reality (XR)

VR hard- and soft-ware



Extended Reality (XR)

VR hard- and soft-ware

VR Applications



Extended Reality (XR)

VR hard- and soft-ware

VR Applications

VR demos





Extended Reality (XR)



Extended Reality (XR)

==

Virtual Reality **U** Augmented Reality **U** Mixed Reality
(VR) (AR) (MR)



Spatial Computing

Extended Reality (XR)

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Virtual Reality (VR) $\cup$ Augmented Reality (AR) $\cup$ Mixed Reality (MR)

Human Computer Interaction timeline

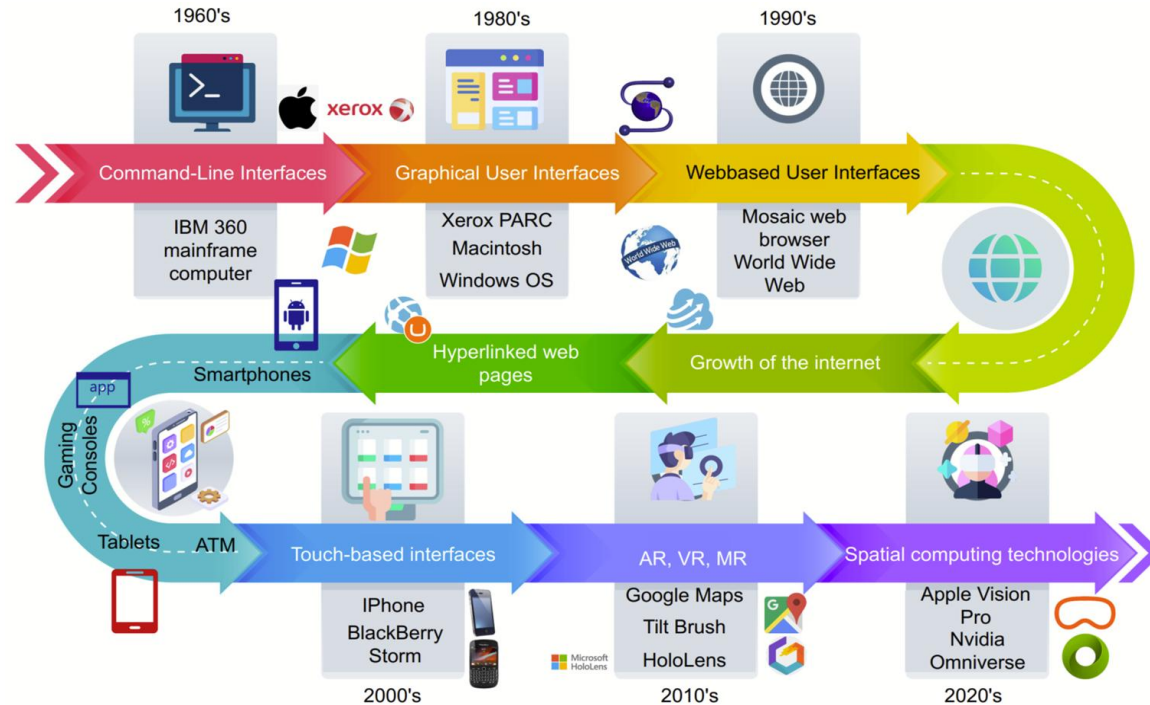


Figure 1: Road Map of User Interfaces.

Human Computer Interaction timeline

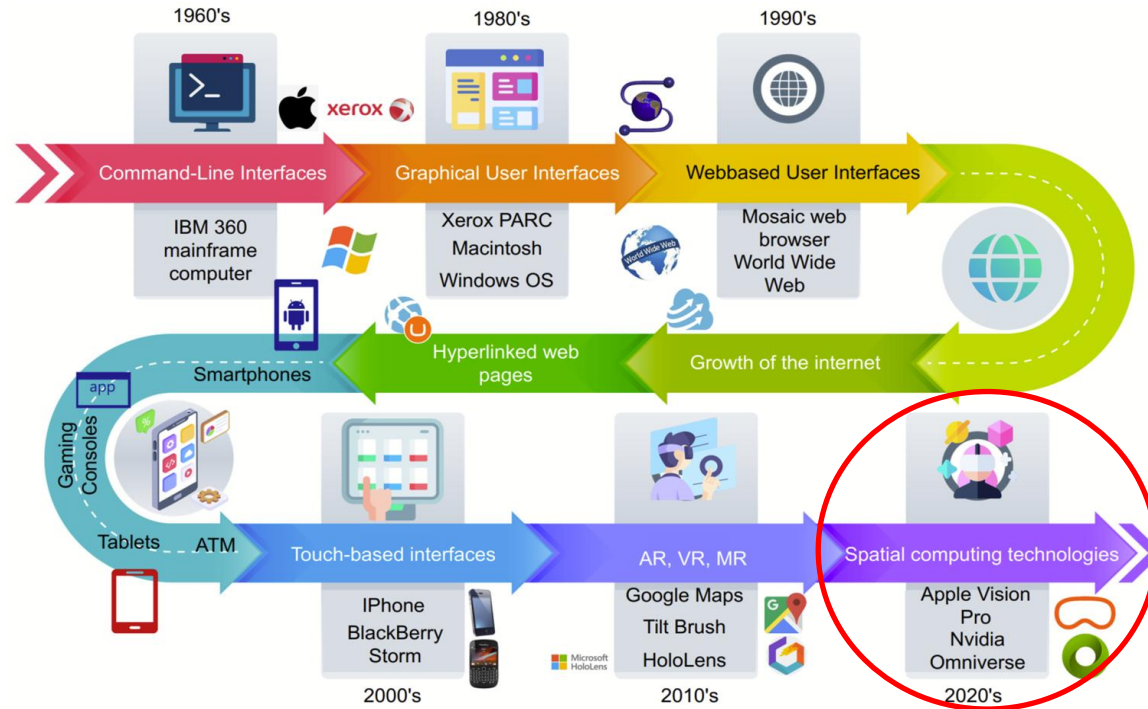


Figure 1: Road Map of User Interfaces.



VR?



VR?

fully immersive and multi-sensory digital experience of a virtual world through a VR system



VR?

fully immersive and multi-sensory digital experience of a virtual world through a VR system

virtual world/world simulation



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virtual world/world simulation





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virtual world/world simulation



user
interaction:
Inclusive
Extensive
Surrounding
Vivid





VR?

fully immersive and multi-sensory digital experience of a virtual world through a VR system

virtual world/world simulation



use of specialized hardware and software: input devices and output devices

user
interaction:
Inclusive
Extensive
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Vivid



VR?

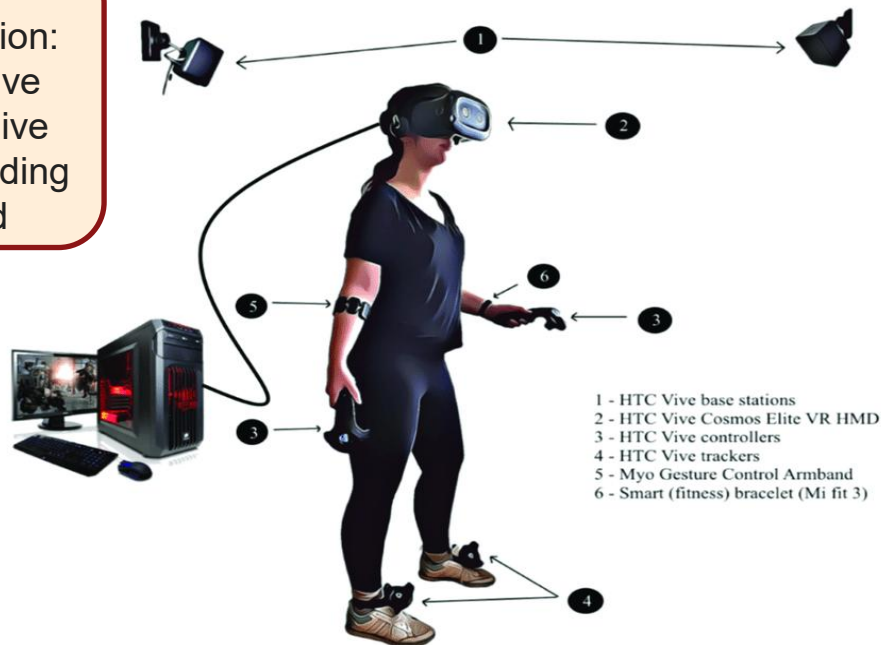
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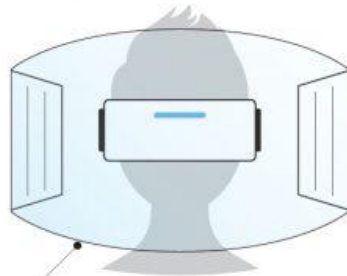
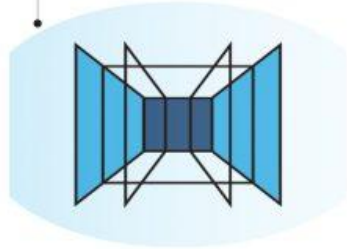
user
interaction:
Inclusive
Extensive
Surrounding
Vivid

use of specialized hardware and software: input
devices and output devices



VIRTUAL REALITY

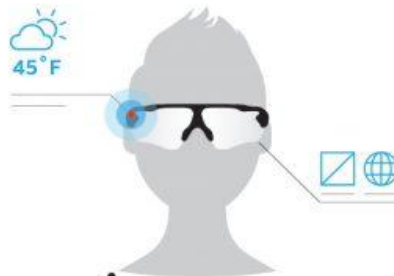
Completely digital environment



Fully enclosed, synthetic experience with no sense of the real world.

AUGMENTED REALITY

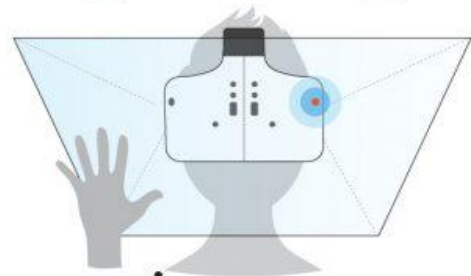
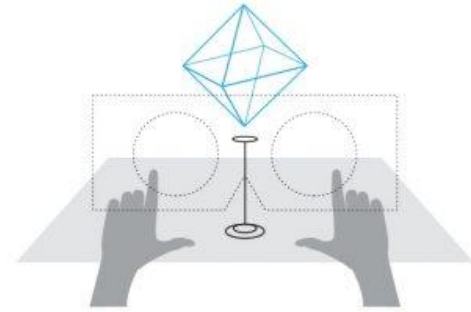
Real world with digital information overlay



Real world remains central to the experience, enhanced by virtual details.

MIXED REALITY

Real and the virtual are intertwined



Interaction with and manipulation of both the physical and virtual environment.

VR hardware

Input devices



VR hardware



Input devices



VR hardware

Output devices





VR hardware/input devices



VR hardware/input devices

record user interactions with objects and environment using sensors



VR hardware/**input devices**

record user interactions with objects and environment using sensors

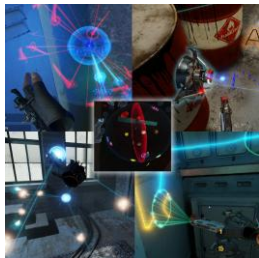


head and hand
movements
based interactions



VR hardware/**input devices**

record user interactions with objects and environment using sensors



head and hand
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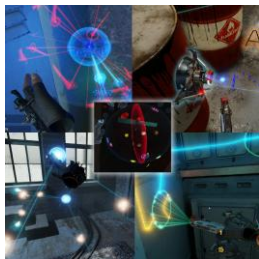


finger based
interactions



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record user interactions with objects and environment using sensors



head and hand
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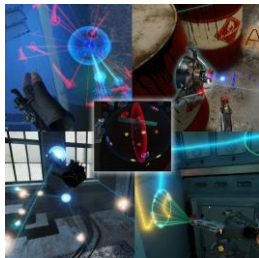


environment
interactions



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environment
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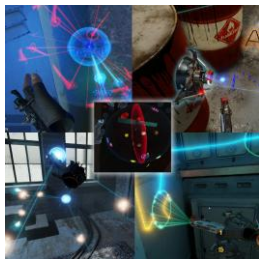
object interactions



VR hardware/**input devices**

record user interactions with objects and environment using sensors

A wide range of input devices characterized by:



head and hand
movements
based interactions



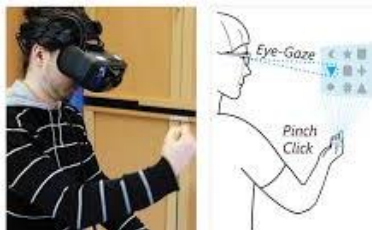
finger based
interactions



environment
interactions



object interactions



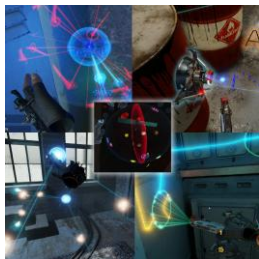
eye-based
interactions

VR hardware/**input devices**

record user interactions with objects and environment using sensors

A wide range of input devices characterized by:

- Accuracy



head and hand
movements
based interactions



finger based
interactions



environment
interactions



object interactions

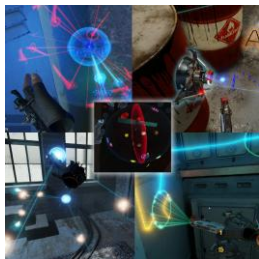


eye-based
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VR hardware/input devices

record user interactions with objects and environment using sensors



head and hand
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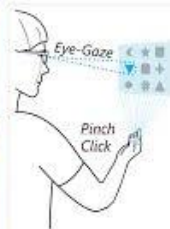
finger based
interactions



environment
interactions



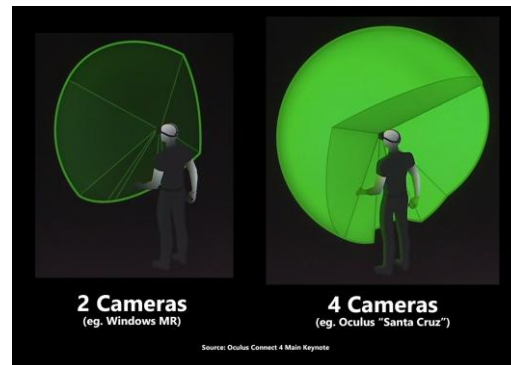
object interactions



eye-based
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A wide range of input devices characterized by:

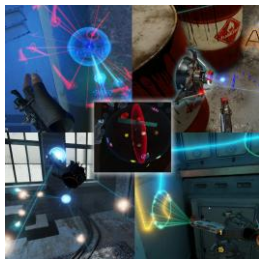
- Accuracy
- Range





VR hardware/input devices

record user interactions with objects and environment using sensors



head and hand
movements
based interactions



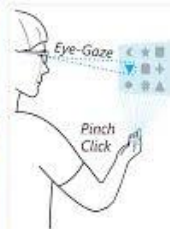
finger based
interactions



environment
interactions



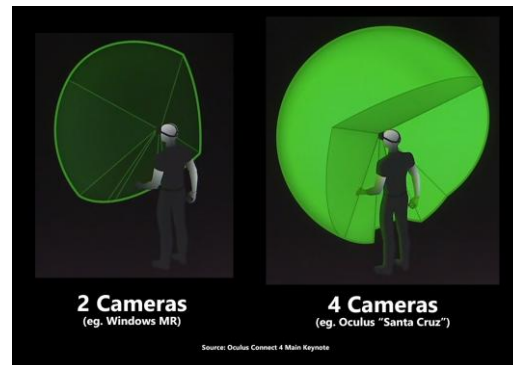
object interactions



eye-based
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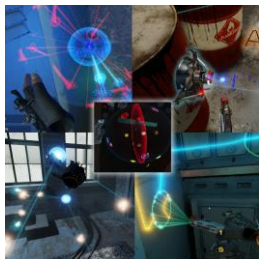
- Accuracy
- Range
- One-time event generation vs continuous stream of event generation





VR hardware/input devices

record user interactions with objects and environment using sensors



head and hand
movements
based interactions



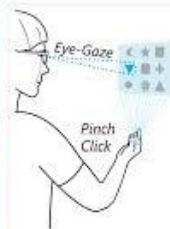
finger based
interactions



environment
interactions



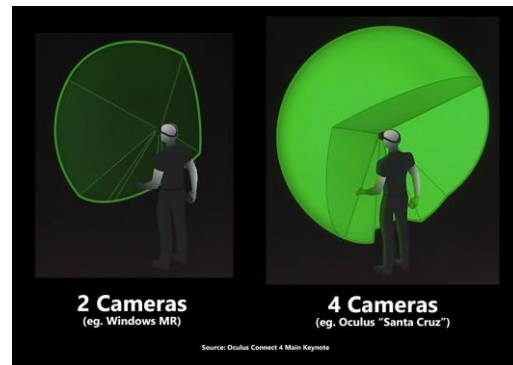
object interactions



eye-based
interactions

A wide range of input devices characterized by:

- Accuracy
- Range
- One-time event generation vs continuous stream of event generation
- Physical medium





VR hardware/input devices/tracking



VR hardware/input devices/tracking

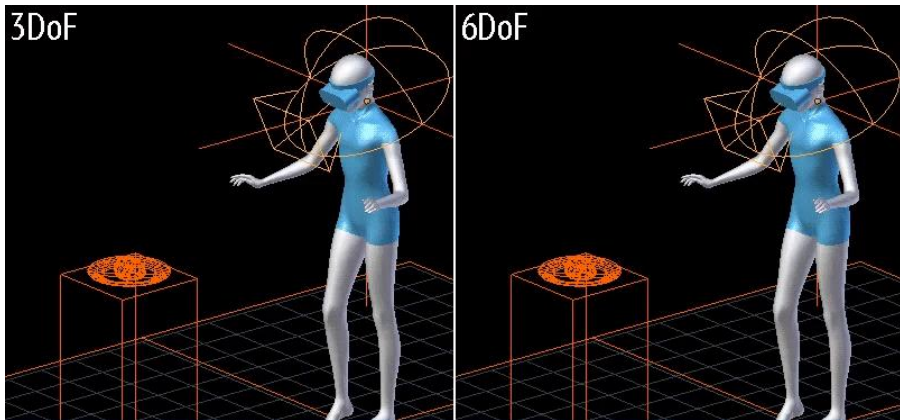
An input device continuously determines the translation and/or rotation in a reference space of a rigid body



VR hardware/input devices/tracking

An input device continuously determines the translation and/or rotation in a reference space of a rigid body

Degree of freedom (DOF)

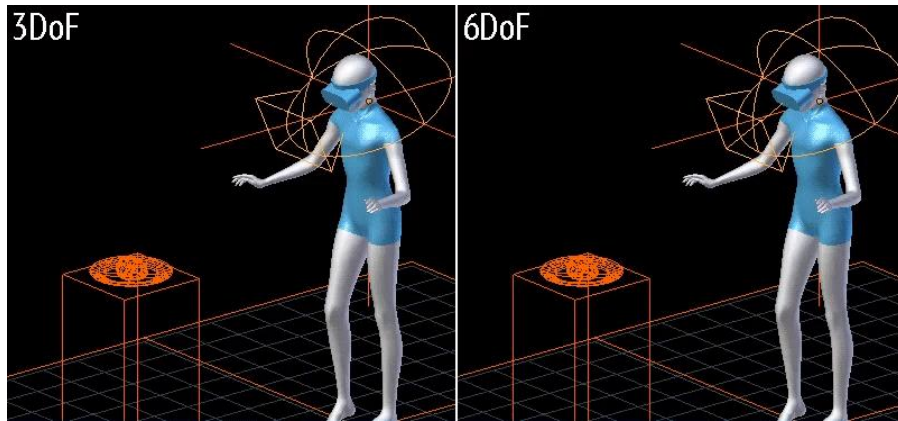




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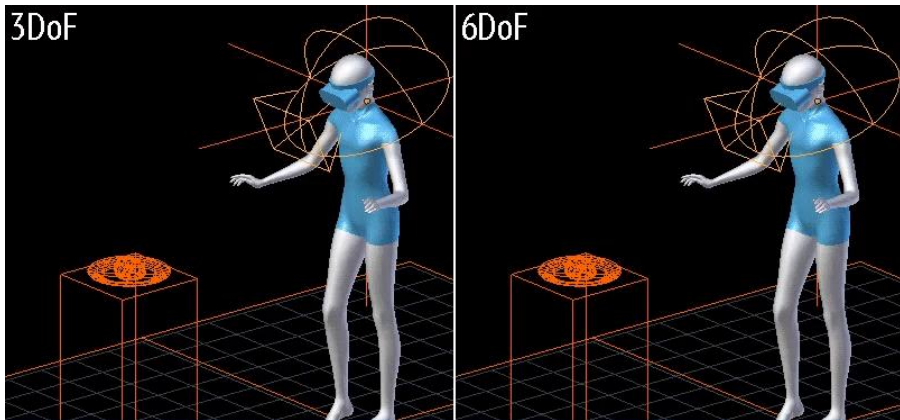




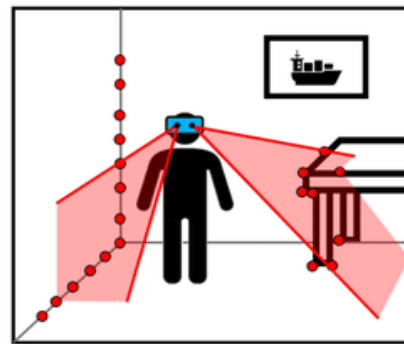
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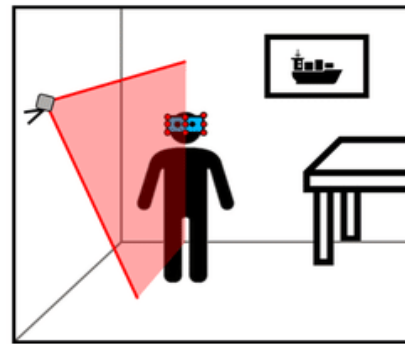
Degree of freedom (DOF)



Inside-out vs Outside-in tracking methodologies



Inside-out tracking (a)



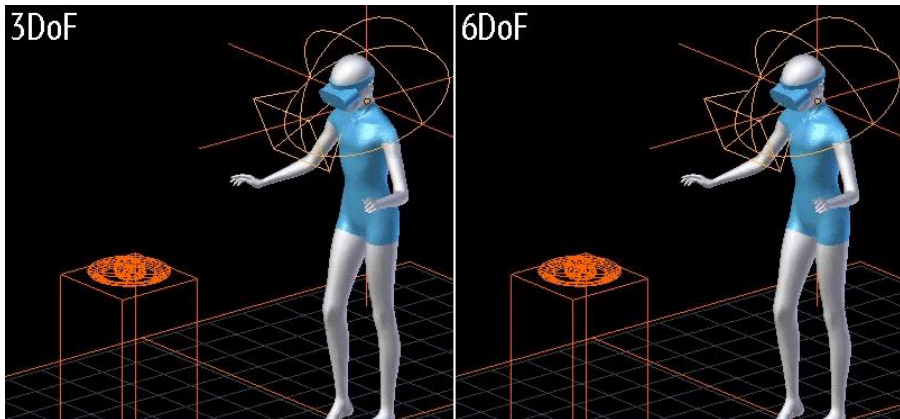
Outside-in tracking (b)



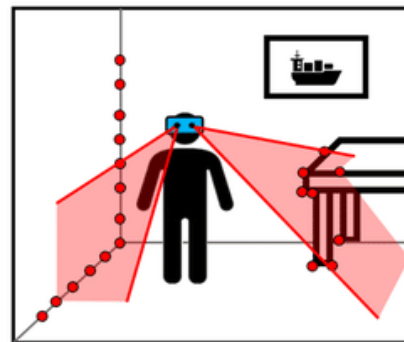
VR hardware/input devices/tracking

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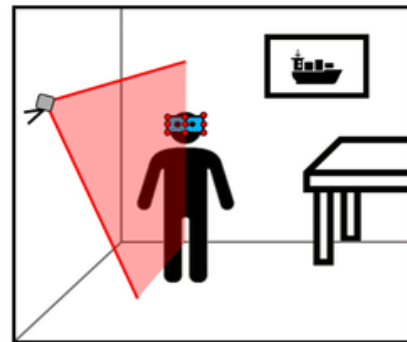
Degree of freedom (DOF)



Inside-out vs Outside-in tracking methodologies



Inside-out tracking (a)



Outside-in tracking (b)



HTC Vive



Oculus Rift



VR hardware/input devices/tracking/techniques

tracking techniques:

acoustic, magnetic field-based, inertial, laser-based, outdoor position tracking



VR hardware/input devices/tracking/techniques

tracking techniques:

acoustic, magnetic field-based, inertial, laser-based, outdoor position tracking

Camera-based tracking



VR hardware/input devices/tracking/techniques

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Camera-based tracking



marker-based methods



VR hardware/input devices/tracking/techniques

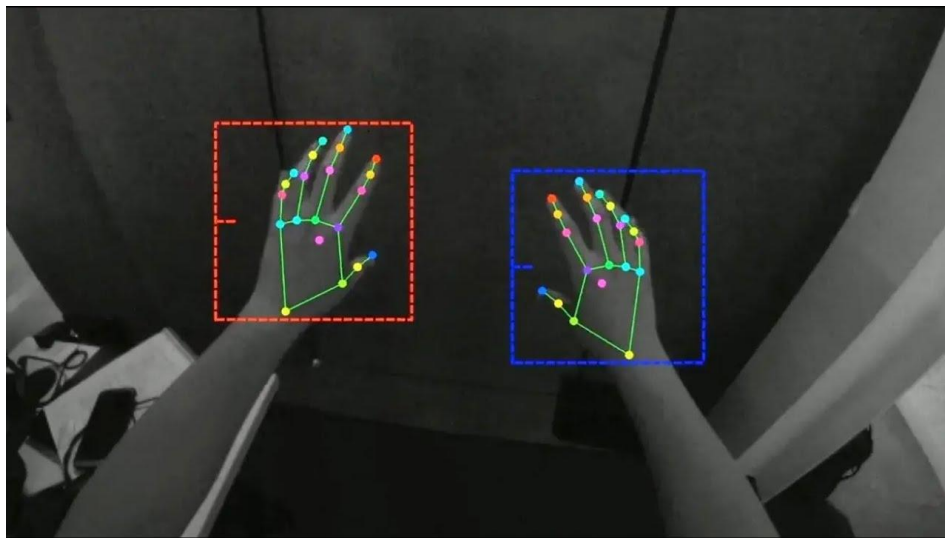
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Camera-based tracking



marker-based methods



feature-based/marker-less methods



VR hardware/input devices/tracking/examples



VR hardware/input devices/tracking/examples

Meta Quest 2



VR hardware/input devices/tracking/examples

Meta Quest 2

Camera-based
hand controller tracking

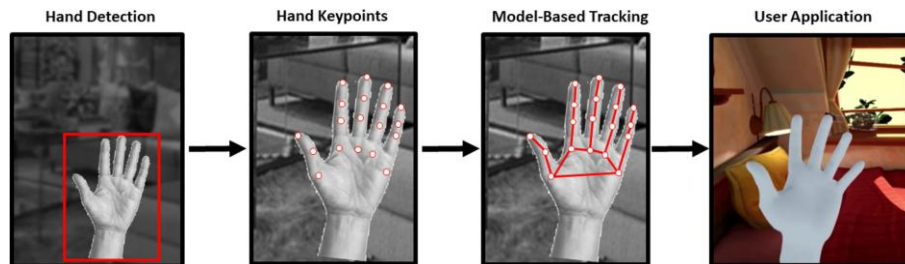
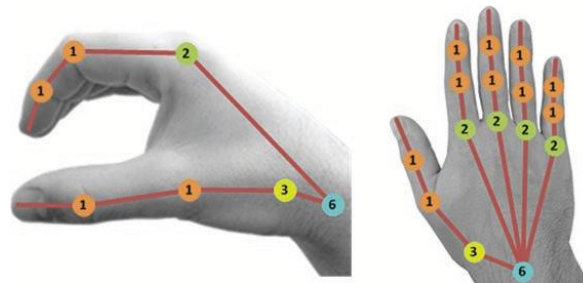


Meta Quest 2

Camera-based
hand controller tracking



Camera-based
hand tracking





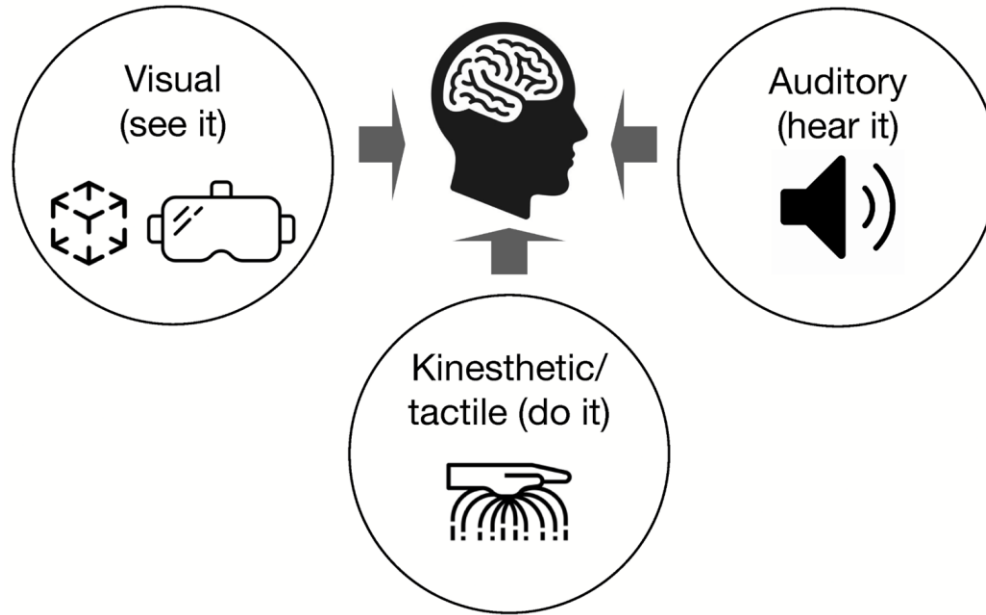
VR hardware/output devices

generate a **representation** of the user interactions with objects and environment
appealing to the **senses** of the users



VR hardware/output devices

generate a **representation** of the user interactions with objects and environment
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VR hardware/output devices/visual output devices



VR hardware/output devices/visual output devices



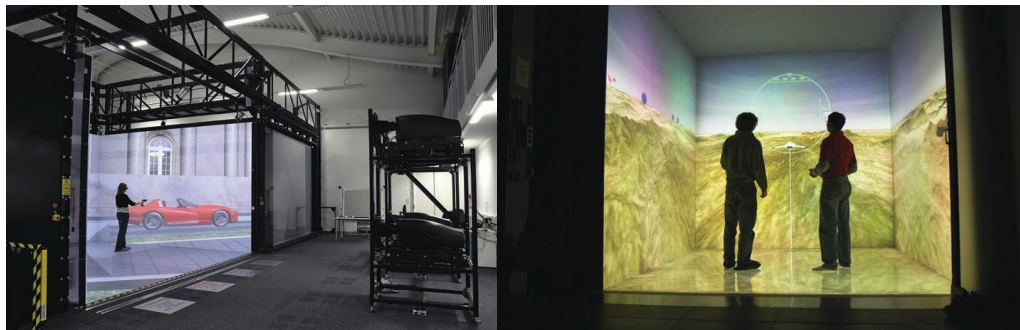
Head-mounted displays



VR hardware/output devices/visual output devices



Head-mounted displays



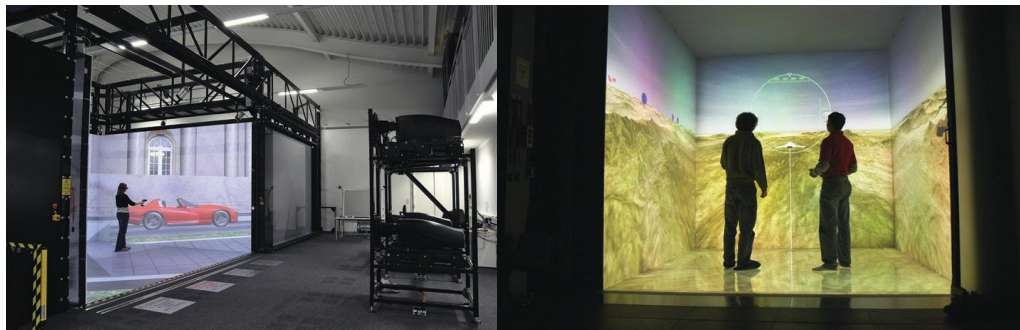
Caves



VR hardware/output devices/visual output devices



Head-mounted displays



Caves



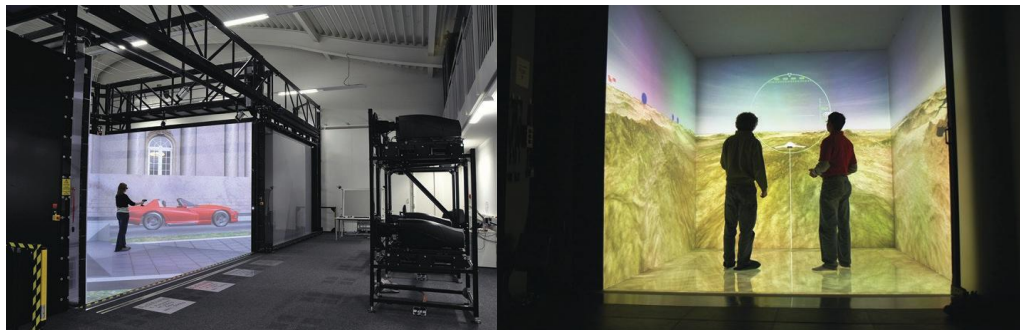
Curved screen projection



VR hardware/output devices/visual output devices



Head-mounted displays



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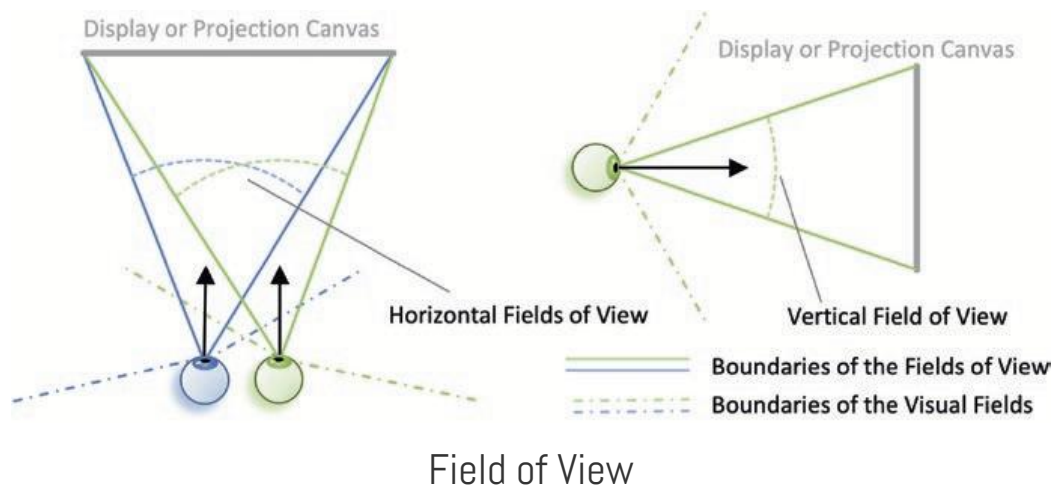


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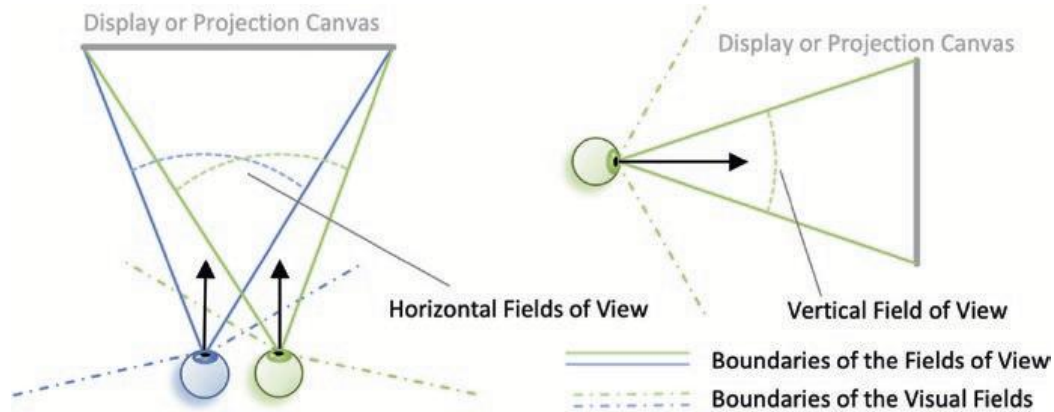


Screen tiled walls

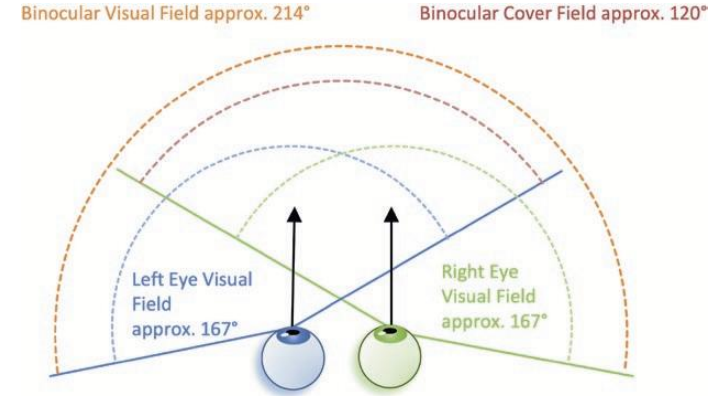
VR hardware/output devices/visual output devices



VR hardware/output devices/visual output devices



Field of View



Visual Field



VR hardware/output devices/visual output devices

Other basic properties of visual output devices:



VR hardware/output devices/visual output devices

Other basic properties of visual output devices:

Frame update rate: describes the resolution of an output device in time



VR hardware/output devices/visual output devices

Other basic properties of visual output devices:

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Latency: time needed to output the transferred data



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VR hardware/output devices/visual output devices

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Interpupillary distance (IPD): stationary and mobile systems



VR hardware/output devices/visual output devices

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Monocular vs binocular: one display vs two displays with associated optics



VR hardware/output devices/visual output devices

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Display technology: LCD, OLED, Micro-OLED...



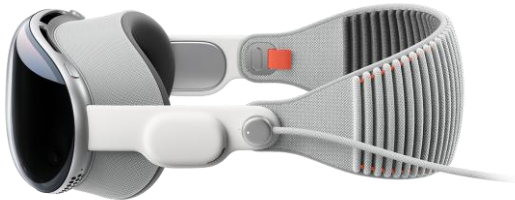
VR hardware/output devices/visual output devices/HMD



VR hardware/output devices/visual output devices/HMD



Google Cardboard



Apple Vision Pro



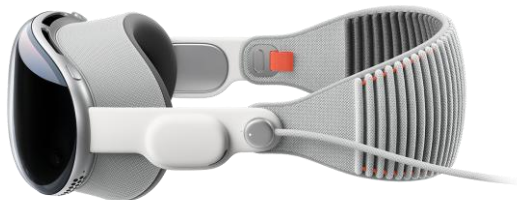
VR hardware/output devices/visual output devices/HMD



Google Cardboard



Varjo X4



Apple Vision Pro



Pimax Crystal



VR hardware/output devices/visual output devices/HMD



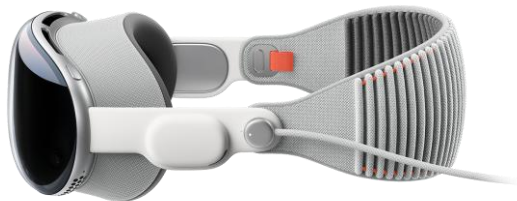
Google Cardboard



Meta Quest 2



Varjo X4



Apple Vision Pro



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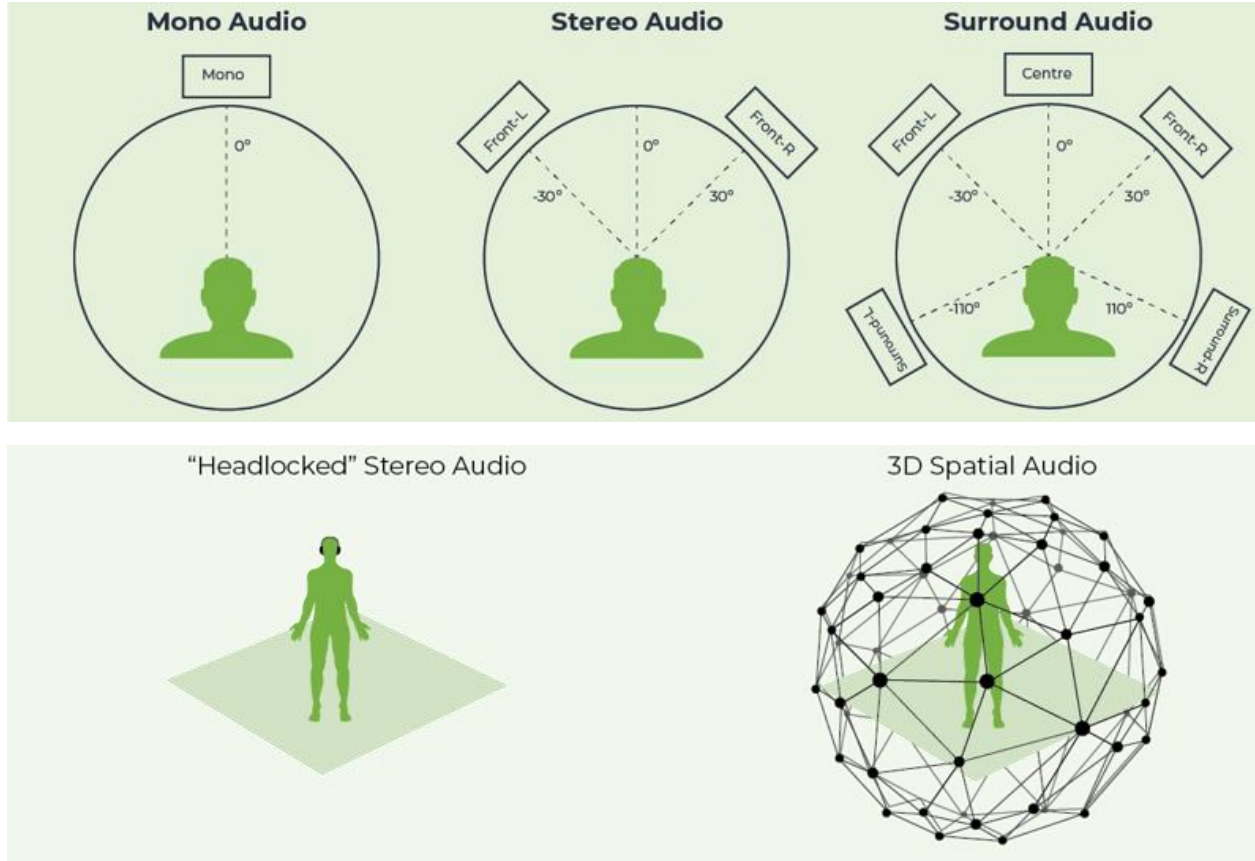


VR hardware/output devices/audio output devices

VR hardware/output devices/audio output devices



VR hardware/output devices/audio output devices





VR hardware/output devices/haptic output devices

tactile feedback



force feedback





Info	Manufacturer	Varjo	HTC	Oculus	Apple	Meta
	Website	varjo.com	vive.com	oculus.com	apple.com	meta.com
	Device Type	PC-powered VR	Standalone VR	Standalone VR	Standalone VR includes external battery	Standalone VR
	Platform	SteamVR	Viveport	Oculus Home, SteamVR Facebook account required	visionOS	Meta Quest
	Announced	December 1, 2020	January 5, 2023	September 16, 2020	June 5, 2023	June 1, 2023
	Release Date	January 1, 2021	March 31, 2023	October 13, 2020	February 2, 2024	October 10, 2023
	Retail Price	\$3195 headset only	\$1099 with controllers	\$299 with controllers. 64 GB model: \$299 (discontinued), 256 GB model: \$349	\$3499 headset only 256GB model: \$3700, 1TB: \$3900	\$499 with controllers 128 GB model: 512 GB model: \$649
Optics	Optics	Aspherical lenses	Pancake lenses	Fresnel lenses	Pancake lenses	Pancake lenses
	Ocularity	Binocular	Binocular	Binocular	Binocular	Binocular
	IPD Range	59–71 mm hardware adjustable (automatic)	54–73 mm hardware adjustable (manual)	58–68 mm hardware adjustable (manual) 3 fixed IPD settings: 58mm, 63mm, 68mm	51–75 mm hardware adjustable (automatic)	58–71 mm hardware adjustable (manual)
	Adjustable Diopter	✗	✓	✗	✗ Prescription inserts available	✗ Prescription lens inserts sold separately
	Passthrough	Passthrough via tracking cameras	16MP RGB camera	Grayscale via tracking cameras	Dual 6.5MP passthrough cameras	Dual 18 PPD color passthrough cameras
Display	Display Type	2 x LCD binocular *Peripheral displays. Focal area: uOLED displays.	2 x LCD binocular	Single Fast switch LCD binocular	2 x Micro-OLED binocular	2 x LCD binocular
	Subpixel Layout	 RGB stripe 3 subpixels per pixel	 RGB stripe 3 subpixels per pixel	 RGB stripe 3 subpixels per pixel	 RGB π type 3 subpixels per pixel	 RGB stripe 3 subpixels per pixel
	Peak Brightness	100 nits				
	Resolution	2880x2720 per-eye *Peripheral displays. Focal displays: 1920x1920	1920x1920 per-eye	1832x1920 per-eye	3660x3200 per-eye estimate	2064x2208 per-eye
	Refresh Rate	90 Hz	90 Hz	120 Hz	100 Hz Also supports 96 Hz and 90 Hz modes	120 Hz



VR software



VR software (development)/OpenXR

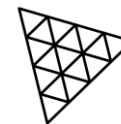
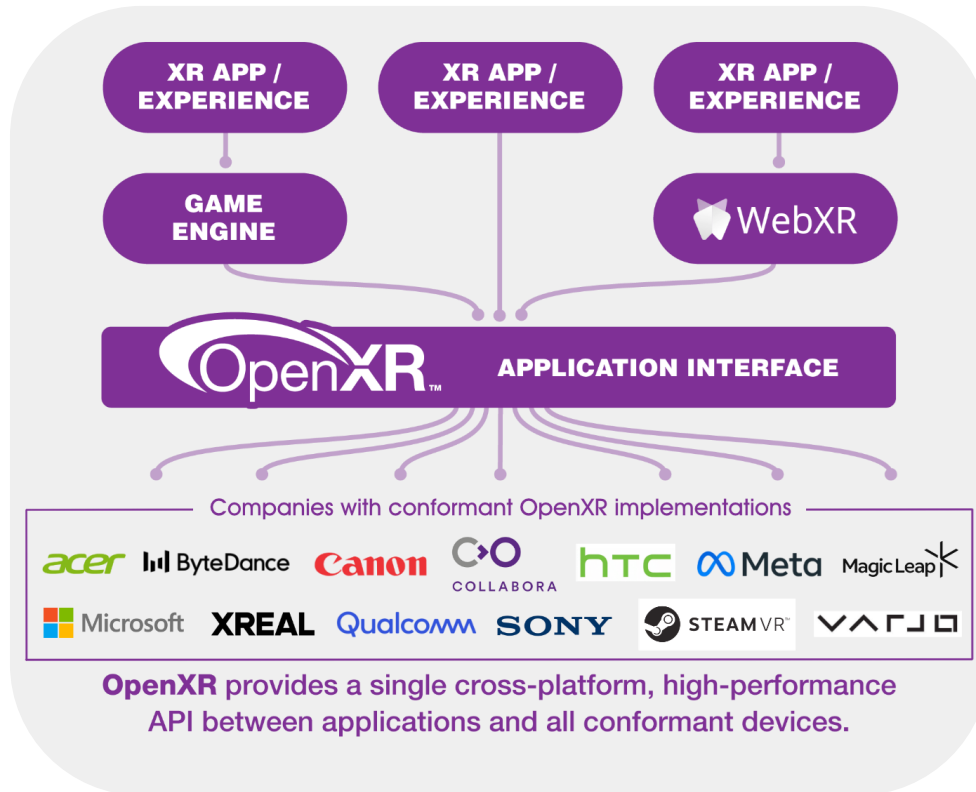


VR software (development)/OpenXR





VR software (development)/OpenXR



three.js

from [OpenXR@KhronosGroup](https://openxr.khronos.org/)



VR applications



VR applications

Training and Education

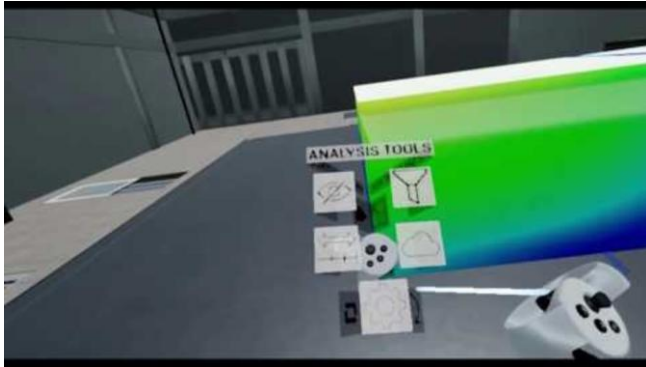
Cultural heritage

Collaborative drug design and visualization

Entertainment



VR applications



Training and Education

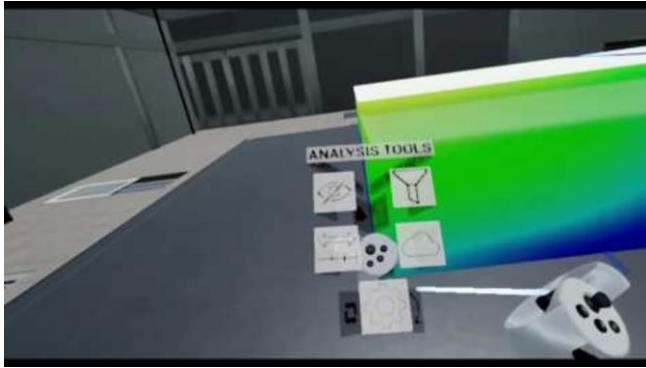
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VR applications



Training and Education



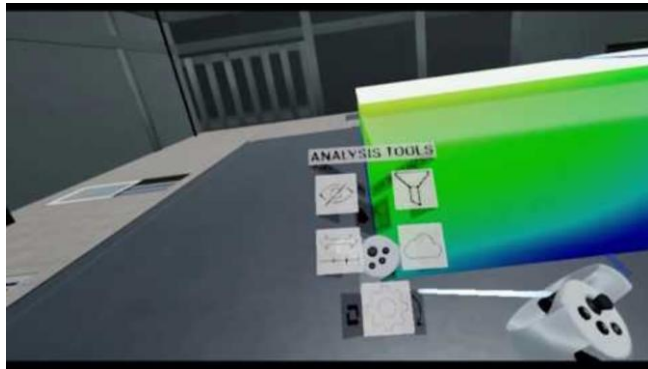
Cultural heritage

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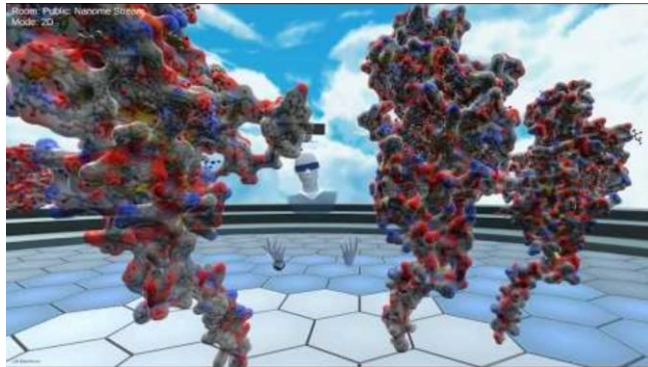
VR applications



Training and Education



Cultural heritage

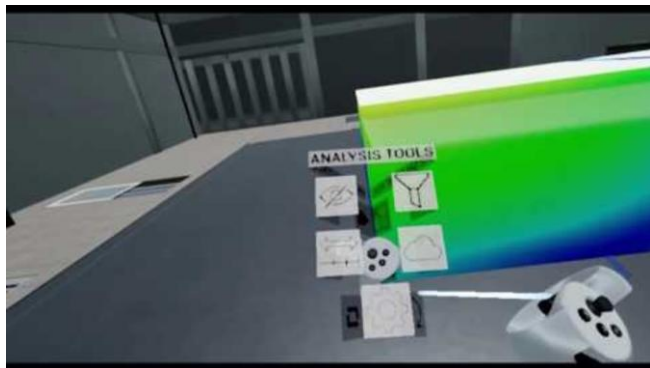


Collaborative drug design and visualization

Entertainment



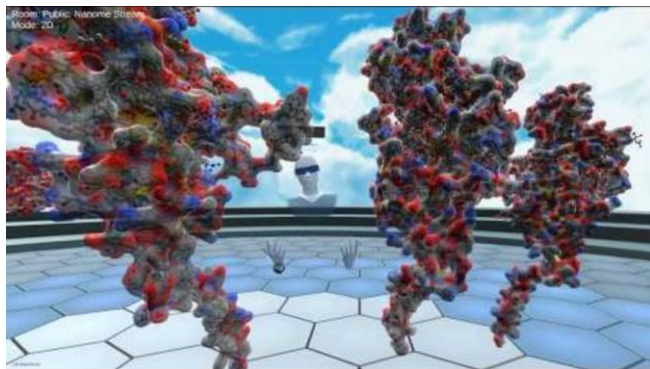
VR applications



Training and Education



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Entertainment

VR demos

